

1.3 Linear Combinations and Span

In the previous section, we studied subspaces. A natural question arises: given a set of vectors, how can we describe the “smallest” subspace that contains them? This leads us to the concept of linear combinations and the span.

1.3.1 Linear Combinations

Definition 1.3.1 — Linear Combination

Let V be a vector space over F , $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then the expression

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_m\mathbf{v}_m$$

is called a **linear combination** of S . The scalars a_i are called the **coefficients** of the linear combination.

Example. (Examples of Linear Combinations)

Below are some examples of linear combinations in different vector spaces:

1. The vector $(2, 1, 0)$ in $\mathbb{R}^3$ is a linear combination of $S = \{(1, 2, 3), (4, 5, 6)\}$ because

$$(2, 1, 0) = -2 \cdot (1, 2, 3) + 1 \cdot (4, 5, 6).$$

2. The polynomial $f(x) = x^3 + 2x + 1$ in $\mathbb{R}[x]$ is a linear combination of $S = \{1, x, x^2, x^3\}$ because

$$f(x) = 1 \cdot 1 + 2 \cdot x + 0 \cdot x^2 + 1 \cdot x^3.$$

1.3.2 Span and Spanning Sets

Definition 1.3.2 — Span

Let S be a subset of a vector space V over F . The **span** of S , denoted by $\text{span}_F(S)$, is the set of all linear combinations of finite subsets of elements of S .

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then $\text{span}_F(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n : a_i \in F\}$. By convention, $\text{span}_F(\emptyset) = \{\mathbf{0}\}$.

Remark. When the underlying field is clear, we abuse the notation of span_F and write it as span .

Theorem 1.3.1 — Spanning Set as Subspace